Project Proposal

CSE312: Database Management System Lab

Submitted To

Syed Eftasum Alam (SEA)
(Lecturer)
Department of CSE, Daffodil International University

Team Members

No.	Name	Student Id
1.	Md. Emon Hossain (**TL**)	232-1-818
2.	Mostafizur Rahman Antu	241-15-631
3.	Nahid Al Galib	241-15-577
4.	Mahjabin Sultana	241-15-899
5.	Nadira Tanzim	241-15-397

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MediSeba

MediSeba

DIGITAL CLINIC

Your Health, Your Care, Anywhere

“Online Doctor Appointment, Payment & Prescription Management System”

1. Introduction

1.1 Overview of the Project Topic

"MediSeba" is a database-driven healthcare system that helps patients book doctor appointments online, receive a token number, complete payments, and store prescriptions digitally. The system supports three roles: **Admin**, **Doctor**, and **Patient**.

1.2 Background of the System

Many clinics and diagnostic centers still manage appointments and prescriptions manually. This creates problems such as:

- Token/serial conflicts when many patients request appointments on the same day
- Difficulty tracking payment status (pending/success/failed/refunded)
- Prescriptions being lost or not easily accessible for future visits
- Patient medical details (allergies, chronic conditions, emergency contact) not being stored in a structured way

A centralized system is required to reduce these issues and provide a faster, more reliable service.

1.3 Importance of Database Systems in Solving These Problems

A relational database system is essential because it ensures:

- **Data integrity** using primary keys, foreign keys, and constraints
- **Consistency** (prevents duplicate token numbers for the same doctor and date)
- **Efficient retrieval** of records for appointments, payments, and prescriptions
- **Better security and management** through role-based access and verified user data
- **Scalability** to support more doctors, patients, and appointments over time

1.4 Overview of the Proposed Solution

The proposed system will:

- Register users using phone number and verify with **OTP**
- Maintain doctor profiles and weekly schedules (day, time, max patients)
- Allow patients to book appointments with auto-managed token numbers
- Track payments with transaction IDs and payment statuses
- Store prescriptions linked to appointments (symptoms, diagnosis, medicines, advice, follow-up date)

2. Problem Statement

Manual appointment and token management often leads to **serial conflicts**, inefficient service, and errors in record-keeping. Payment tracking becomes difficult, and prescriptions are not stored safely for future reference. Patient health information is not organized, which reduces service quality.

Therefore, a structured database-based system is required to manage appointments, payments, and prescriptions accurately and efficiently.

3. Scope of the Project

3.1 Features Included in the System

A) User & Verification

- User registration and login (Admin/Doctor/Patient)
- Phone-based **OTP verification** (expiry + one-time use)

B) Doctor Management

- Doctor profile: specialization, qualification, experience years, consultation fee, bio, status
- Doctor schedule: day of week, start time, end time, maximum patients per schedule

C) Patient Profile Management

- Personal details: DOB, gender, blood group, address
- Health details: allergies, chronic conditions
- Emergency contact details and profile photo

D) Appointment & Token Management

- Appointment booking with appointment date
- Token number tracking per doctor per date
- Appointment status tracking: confirmed / in_progress / done / cancelled / no_show

E) Payment Management

- Payment status: pending / success / failed / refunded
- Transaction ID storage (unique)
- Refund time tracking

F) Prescription Management

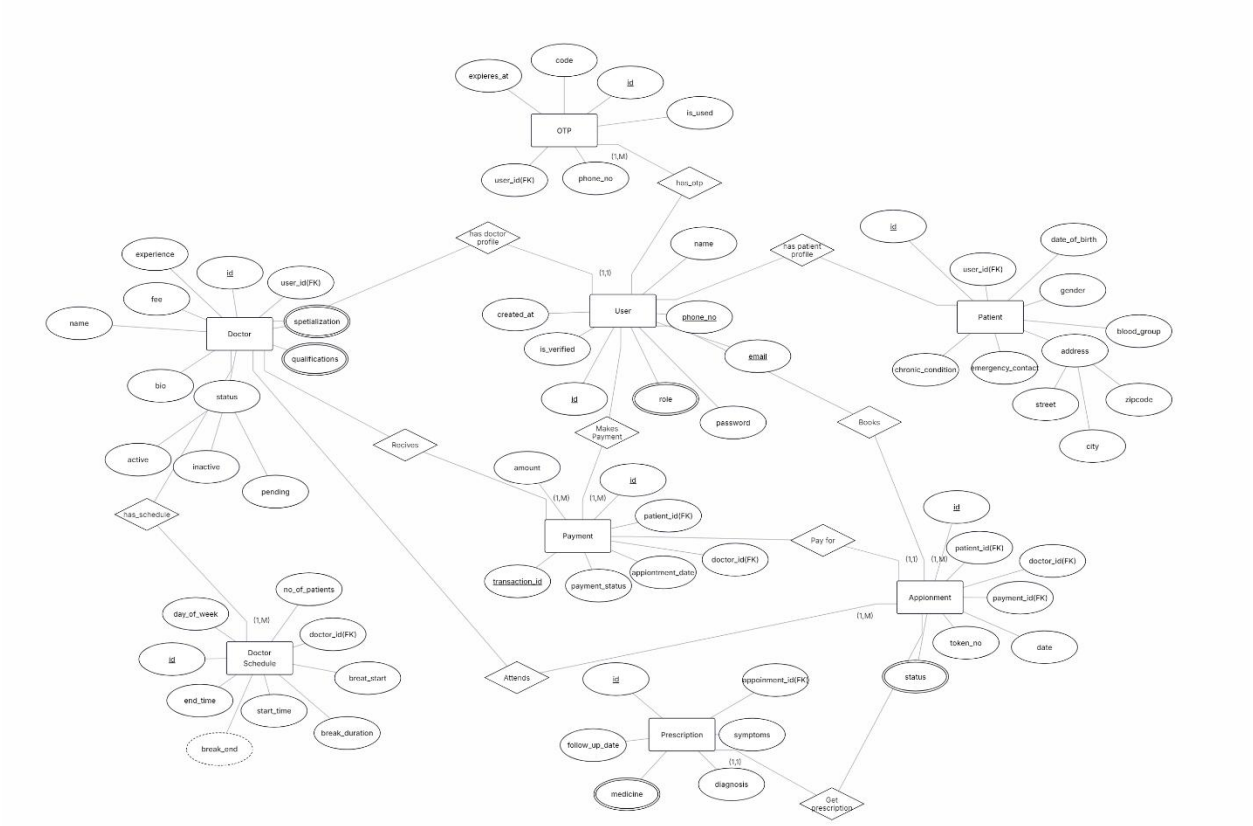
- Prescription creation linked to appointment
- Store symptoms, diagnosis, medicines, advice, follow-up date

3.2 Target Users

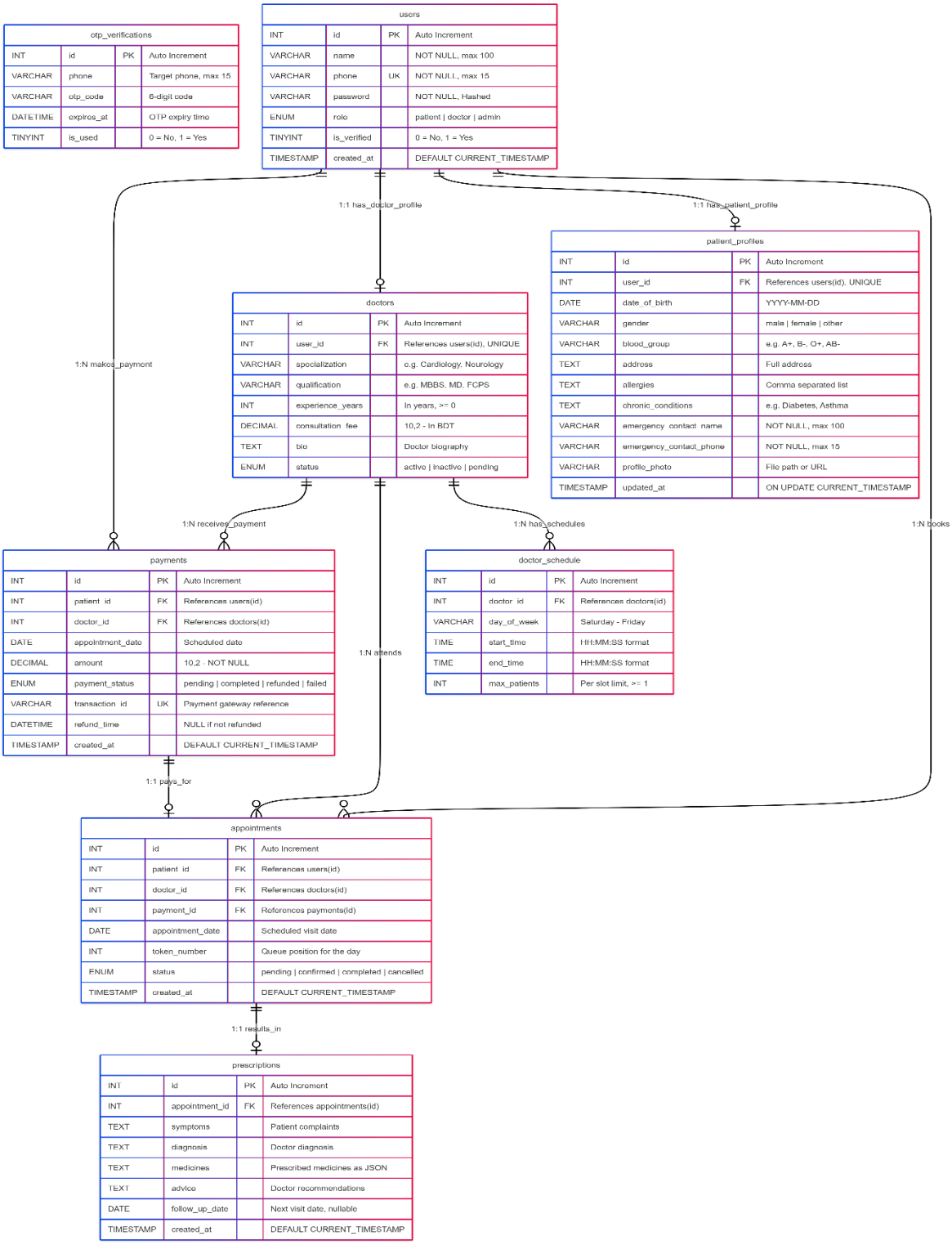
- **Admin:** Manage users and monitor appointments, payments, and overall activities
- **Doctor:** Manage schedules, view appointments, update appointment status, create prescriptions
- **Patient:** Maintain profile, book appointments, pay fees, view prescriptions and follow-up information

4. Database Design (ER-Diagram) 2NF

4.1 Chen Notation Diagram:



4.2 Mermaid Notation Diagram:



5. Tools

5.1 Front-end Tools

- **HTML5** – Web page structure
- **CSS3** – Design, layout, and responsiveness
- **JavaScript** – Form validation and dynamic interactions

5.2 Back-end Tools

- **PHP** – Server-side logic, role-based access, appointment and payment processing

5.3 Database & Server Tools

- **XAMPP** – Local server environment (Apache + MySQL + PHP)
- **MySQL** – Relational database system
- **phpMyAdmin** – Database management and query execution tool

6. Conclusion

MediSeba provides a complete, database-centered solution for managing doctor appointments, token numbers, payments, and prescriptions. By using a relational database with proper constraints and normalization, the system improves reliability, reduces conflicts, and ensures structured storage of patient and medical records.